From resource to risk:

Generation, demand and residual load in a renewable Europe

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Introduction

The energy transition:

Powering Europe with wind & sun

Opportunity, variability, and the challenge of reliability

42.5%
binding RE target by 2030
(up from 25% in 2024)

1.5°C
Paris Agreement
warming target

Net zero
EU target
by 2050

The variability challenge

- Wind and solar produce according to *meteorological conditions* – not according to demand.
- Output can drop sharply within hours, especially for individual regions during calm or cloudy periods.
- Low-generation events can persist for days and affect entire regions simultaneously.

This hackathon quantifies all three at hourly resolution across Europe, 2012–2014.

Adequacy
How often and how severely can demand exceed supply?

Flexibility
How large and frequent are critical ramp events?

Security
What do the extremes look like, and how bad can they get?

Understanding a renewable power system requires three lenses

Renewable supply

Variable, weather-driven.
Where is the resource?
How severe can low-generation spells be?

$-$ →

Electricity demand

Driven by human activity.
When are peaks?
How does it vary spatially?

$=$ →

Residual load

Demand minus renewables.
What must dispatchable capacity always cover?

This hackathon tackles all three – using the RE-Europe dataset (Jensen & Pinson, 2017)

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Motivation & context

The RE-Europe dataset

Power systems built around wind and solar couple tightly to **weather and climate dynamics**, yet open models rarely capture this coupling at continental scale.

Key barriers:

- Confidentiality & commercial interests limit data sharing
- Multi-disciplinary gap: meteorology $\times$ economics $\times$ power engineering
- Existing open datasets cover only *parts* of the problem
- No common benchmark for the full European system

Objective

A **common benchmark**: complete, open-access, ready-to-use for the full European electricity system.

Foreseen uses

Market design studies; cross-zonal coordination; renewable forecasting; network analysis.

Jensen & Pinson (*Scientific Data* 4:170175, 2017) — the first open dataset combining network, generators, demand and weather-driven renewables for mainland Europe.

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Dataset overview

Dataset at a glance Six components: three static, three dynamic

Component	Coverage / Size	Resolution
Static components		
Transmission Network	1,494 buses, 2,156 lines	Continental Europe
Generator Database	969 units, 9 fuel types	Individual plant
Capacity Layouts	Uniform & proportional	Per bus
Dynamic components — 3 years, hourly, per bus		
Demand Signal	3 years (2012–2014)	Hourly, per bus
Wind & Solar Signal	3 years (2012–2014)	Hourly, per bus
Wind & Solar Forecast	91 h ahead, issued 00:00 & 12:00	Hourly, per bus

Groups 1 & 2 (wind & solar): wind & solar signals and capacity layouts · **Groups 3 & 4** (demand): demand signal · **Groups 5 & 6** (residual load): all dynamic components

Component details

Transmission network Hutcheon & Bialek (2013) model, georeferenced

- **1,494 buses** connected by **2,156 AC lines** + 1 HVDC link; voltage levels 110–380 kV
- Covers **mainland Europe**, excluding British Isles, Fennoscandia and Baltic states
- Typical line length 30–70 km: well-matched to the **spatial scale of weather systems** driving renewable variability
- Bus coordinates georeferenced with RMS error ≈ **17 km**: sufficient for hourly renewable dynamics

The network defines *where* each signal lives: all wind, solar, demand and residual load signals are anchored to its 1,494 nodes.

Network files

`network_nodes.csv`	1,494 rows
`network_edges.csv`	2,156 rows
`network_hvdc.csv`	1 row

Key design choice

Network granularity determines the *spatial resolution* of all other dataset components: finer than country level, coarser than individual plants.

Relevant for: All groups: node coordinates define spatial structure of every signal; Groups 5 & 6: line limits bound how residual load can be redistributed

Generator database Based on Global Energy Observatory; 969 units, 9 fuel classes

Coverage & parameters

- Source database covers **78%** of European installed capacity (2012)
- Location, fuel type and nameplate capacity per unit
- **Marginal costs** estimated by fuel type from EIA 2014 data; ±10% random variation added to reflect real-world cost diversity

Conventional generators are the backup when renewables fall short: their capacity and cost structure define the floor of residual load that must always be covered.

Marginal generation costs

Fuel type	Cost [$/MWh]
Nuclear	lowest
Lignite	∼low
Coal (IGCC)	mid
Natural Gas (CC)	mid–high
Fuel Oil	highest

Note: CO_2 costs *not* included. Users are encouraged to substitute their own cost estimates. Relevant for: Groups 5 & 6: defines dispatchable capacity available when renewables fall short.

Renewable energy signals & forecasts

Two independent meteorological sources

- **COSMO-REA6** reanalysis (≈7 km, hourly) → the "observed" production signal you will use for analysis
- **ECMWF** deterministic forecast (≈16 km, issued every 12 h for 91 h ahead) → forecast signal, consistent with ECMWF reanalysis
- Two sources give a **built-in sensitivity check**: results should be robust across both
- Signals are **capacity factors** (0–100%): multiply by a capacity layout to get MW

Physical conversion uses a Siemens SWT-107 wind turbine power curve and a Scheuten P6-54 solar panel model, with wind speeds interpolated to 80 m hub height.

Seam smoothing (wind)

Stitching 12-hourly ECMWF forecasts creates unphysical jumps. These are smoothed out via interpolation over 1 h at each seam: after smoothing, typical wind speed changes are within the normal range.

Solar: simpler fix

First **24 h** of each midnight forecast used: no solar radiation at night means no jump at the seam.

Groups 1 & 2: primary data source
Groups 5 & 6: wind & solar signals drive residual load

Demand signal & capacity layouts

Demand disaggregation

- Country-aggregate hourly load from **ENTSO-E**, distributed to buses proportional to **population density**
- Total country load preserved at every hour
- The **fraction of national load assigned to each bus is fixed**: demand fluctuates in time but its spatial distribution does not.

Relevant for: Groups 3 & 4: primary data source; Groups 5 & 6: demand drives residual load.

Two capacity layouts

Uniform

Capacity $\propto$ area per bus.
Even spread across Europe.

Proportional

Capacity $\propto$ area $\times$ mean capacity factor.
More capacity in high-resource zones.

Both scaled so that total wind or solar production matches total EU load over 3 years: i.e. **100% gross penetration**. Scale by $\alpha \in [0, 1]$ to simulate lower penetration levels.

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Validation

Technical validation How good is the data?

Network & wind signal

- Georeferencing RMS error $\approx$ **17 km**: acceptable for hourly renewable dynamics
- Wind seam smoothing reduces unphysical jumps to within the **normal range** of hour-to-hour wind speed changes

System-level check

- A basic power system simulation confirmed that the generator database contains **sufficient capacity to always meet demand**: the dataset is internally consistent

Comparison with ENTSO-E 2015

- Wind and solar duration curves show **reasonable qualitative agreement** with real-world data
- Spain wind: dataset shows more low-production hours than reality: likely a meteorological bias at low wind speeds
- Spain solar: ENTSO-E data inflated by concentrating solar (CSP), which is not modelled here

Validation is qualitative: the dataset is not designed to replicate real-world signals exactly, but to capture the key spatiotemporal structures of renewable generation and demand across Europe.

Computing residual load

Computing residual load

What must conventional generation always cover?

$$R_{t,n} = L_{t,n} - \mathtt{CF}^{\text{wind}}_{t,n}\,\alpha_w\,C^{\text{wind}}_n - \mathtt{CF}^{\text{solar}}_{t,n}\,\alpha_s\,C^{\text{solar}}_n$$

$L_{t,n}$ load [MWh]

$\mathtt{CF}_{t,n}$ capacity factor

C_n layout capacity [MW]

α penetration scaling

Groups 5 & 6 will compute $R_{t,n}$ across all hours and study its distribution, extremes and sensitivity to the scaling factor α.

Scaling

Layouts normalised to **100%** gross penetration. Choose $\alpha \in [0, 1]$:

$$C_n = \alpha \times C^{\text{layout}}_n$$

e.g. $\alpha = 0.5$ gives 50% penetration.

Interpretation

- $R > 0$: demand exceeds renewables: other sources must fill the gap
- $R < 0$: renewables exceed demand: surplus must be stored or curtailed

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Summary

Key takeaways

1. **First complete open dataset** combining network, generators, demand and weather-driven renewables for all of continental Europe.
2. **3 years** of hourly bus-level signals (2012–2014) from two independent meteorological sources: built-in sensitivity check.
3. Validated against ENTSO-E data; generator capacity is **sufficient to always meet demand** at any renewable penetration level.
4. Fully **open-access** (CC-BY 4.0): designed as a common benchmark for the research community.

In this hackathon you will use this dataset to characterise renewable generation, electricity demand and residual load across mainland Europe: from typical behaviour to extreme events.

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Challenge Overview

The challenge Problem statement and objectives

Problem statement

When renewables fall short and demand peaks, what must the system cover? Using three years of hourly European data, characterise the spatiotemporal structure of renewable generation, demand, and residual load, and quantify the extremes that threaten system adequacy.

Objectives

- Groups 1 & 2: Characterise wind & solar variability, seasonal patterns, and low-generation extremes
- Groups 3 & 4: Analyse electricity demand cycles, spatial concentration, and peak timing
- Groups 5 & 6: Quantify residual load and tail risks across penetration levels

The integration challenge Three interlinked components of a renewable power system

How do generation, demand, and residual load behave, and what do their extremes imply for system adequacy?

Component 1: Renewable Supply

Wind & solar are variable, partially predictable, and spatiotemporally structured. Low-generation episodes are critical for system planning.

Component 2: Demand

Driven by human activity & climate, largely *independent* of the renewable resource. Peaks define the moments of greatest exposure.

Component 3: Residual Load

Demand minus renewables. Must be covered by dispatchable generation, storage, and cross-border exchange at every hour.

Hackathon structure

Six groups, three topics: two groups per topic

Groups 1 & 2

Wind & solar generation

Characterise the renewable *resource*: its variability, spatial structure, seasonal patterns, and the depth & duration of low-generation episodes.

Groups 3 & 4

Electricity demand

Characterise the *load*: diurnal and seasonal cycles, spatial concentration, volatility, and the timing of extreme demand events.

Groups 5 & 6

Residual load

Characterise demand minus renewables: typical behaviour, ramp rates, tail risks, and how these evolve as renewable penetration scales up.

Taken together, the three topics reveal the hours of **maximum system stress**, how their frequency and severity scale with renewable penetration, and what operational and infrastructural responses they demand.

Teams

6 self-organised teams
4–6 members each
Mix both cohorts

Task overview: group by group

Key questions each group must address

Wind & Solar

- Seasonal & diurnal profiles
- Spatial maps of mean capacity factor
- Wind–solar temporal complementarity
- Longest simultaneous low-generation streak
- 5th-percentile tail analysis

Demand

- Diurnal & weekday/weekend profiles
- Country-level mean & peak load maps
- Coefficient of variation by country
- 95th-percentile extreme event characterisation

Residual Load

- Duration curves & histogram
- Stress events (above 90th percentile)
- Sensitivity to $1.5\times$, $2\times$, $3\times$ capacity scaling
- Composite stress index & worst-day ranking

All groups are strongly encouraged to use data visualisation techniques presented in the subsequent training course.

Teams & schedule

Teams Composition and structure

Team Formation

- 6 self-organised teams
- 4–6 members per team
- Mix of skills encouraged
- One team lead per group
- Teams need to consist of students from both cohorts

Schedule Hackathon sessions across the Jamboree

Monday 13 Apr

09:00–10:30
Introduction & group allocations

15:00–15:30
Group work

15:30–16:00
Tea & coffee

16:00–17:00
Group work

Tuesday 14 Apr

09:00–10:30
Group work

10:30–11:00
Tea & coffee

11:00–13:00
Group work

14:00–15:30
Group work (prep presentations)

17:30–18:15
Finalise presentations

Wednesday 15 Apr

09:00–10:30
Group presentations (6×10 min)

11:00–11:30
Prize ceremony